SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1520

COURSE OUTCOME COVERED
CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2) Total Marks: 25
Question Count: 5

Code of Conduct
I G. Yashmita (192511032) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G. Yashmita

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

2. An organization wants to store large volumes of backup data securely in the cloud.
   Discuss how cloud storage solutions ensure durability and data redundancy.
   How can encryption and access control protect sensitive backup data?
   Explain the cost advantages compared to traditional storage systems.
   Provide an example of a disaster recovery use case.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real world e-commerce scenario.

Introduction:

E-commerce companies often experience unpredictable traffic spikes during seasonal sales, festive offers and special shopping events. During these periods, thousands or millions of users may access a website simultaneously. If the infrastructure cannot handle the increased demand, the website may become slow or unavailable. Cloud auto-scaling and load balancing help manage these traffic fluctuations efficiently.

Cloud Auto-Scaling:

Cloud auto-scaling is a cloud computing feature that automatically increases or decreases computing resources according to application demand. It adjusts the number of servers or virtual machines instances based on predefined rules and real-time metrics.

Autoscaling provides the following benefits:

Improved performance.

Cost Efficiency.
High availability
Automatic resource management.

Load Balancing:

Load Balancing distributes incoming network traffic across multiple servers. Instead of sending all requests to a single server, a load balancer directs request to available and healthy servers.

Role of Monitoring Tools:

Monitoring tools continuous observes the performance and health of cloud resources. They collect metrics such as CPU utilization, memory usage, network traffic, request count, and response time. Tools such as Amazon cloudwatch and Prometheus with Grafana help administrators identify performance problems.

Real-World E-commerce Scenario:

Consider an online shopping platform during a major festivale sale. Normally, the website serves a moderate number of customers but during the sale, traffic increases significantly.

AutoScale launches additional application servers, while the load balancer distributes customer request across the

available servers.

Conclusion:

Cloud auto-scaling and load balancing work together to provide scalability, reliability and cost efficiency. Monitoring tools make the system responsive to changing demand.

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to the traditional storage systems. Provide an example of a disaster recovery use case.

Introduction:

Organizations generate large amounts of data, including customer records, financial information, application files and business documents. Cloud storage provides scalable, durable and cost-effective backup solutions.

Durability and Data redundancy:

Durability refers to a ability of a storage system to preserve data without loss overtime. Cloud providers design storage systems with fault detection and recovery mechanisms to reduce the risk of losing stored information.

Data redundancy means maintaining additional copies of data so that information remains available if one copy becomes unavailable or corrupted.

Encryption and Access Control:

Encryption protects sensitive information by converting readable data into an encoded form that can only be accessed using the appropriate cryptographic keys. Cloud backup data can be protected through

- Encryption at rest
- Encryption in transit
- Key management.

Access control ensures that only authorized users and services can access backup data. Role based access control (RBAC), Identity and Access management (IAM), and multi-factor authentication (MFA) help prevent unauthorized access.

Cost Advantages over traditional storage:

Traditional systems require organizations to purchase physical storage devices, maintain data centres provide electricity and cooling and replace aging hardware. Cloud storage reduces the need for large upfront investments because organizations can pay according to their storage usage and selected service tier. Storage capacity can be increased without

purchasing additional physical equipment.

Disaster Recovery Use case:

Consider a banking organization that stores customer account records and transaction databases. If its primary data center experience a fire, ransomware attack or hardware failure, the organization may lose access to critical information.

TO prevent this, the bank regularly backs up its databases to cloud storage. The backups are encrypted, protected by strict access policies and replicated to a separate region. The organization should also test its backups regularly to verify that the data can be restored successfully.

Conclusion:

Cloud storage provides durability, redundancy, security, scalability and cost efficiency. Encryption and access control protect sensitive backup data, while geographically separated copies and tested recovery procedures help organizations and maintain business continuity during disasters.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

Introduction:

Mobile applications used across different countries must deliver content quickly and reliably to users in different geographical locations. If all content is served from a single central server, users who are far away may experience high latency and slow loading times. A Content Delivery Network (CDN) solves this problem by distributing cached content through geographically distributed edge servers.

Content Delivery Network (CDN):

A CDN is a distributed network of servers that delivers web content, images, videos, applications files and other resources from locations closer to users. CDN's improve application performance in several ways

Reduced Latency

Faster content delivery

Reduced origin-server workload
Improved availability
Better scalability.

Role of Edge Servers:

Edge servers are located closer to end users than the central origin server. They store cached copies of frequently accessed content and deliver it with lower network delay.

With a CDN, a cached copy of the video can be delivered from an edge server.

Global Availability through Cloud Providers:

Cloud providers support global availability through multiple regions, availability zones, distributed networking and managed CDN services.

Health checks, traffic routing, replication and failover configuration can help direct users toward healthy infrastructure when a regional service becomes unavailable.

Real-World Example: Video Streaming Application:

Consider a real-time video-streaming application used by viewers in India, Europe and North America. Without a CDN, every viewer may need to retrieve

video segments from a central server, increasing latency and server load.

Conclusion:

CDN's improve global application performance by reducing latency, distributing content through edge servers and reducing the load on origin servers.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring and encryption play in security? How can the company ensure data sovereignty requirements are met?

Introduction:

Financial institutions handles sensitive information such as customer identities, bank account details, credit card data and financial transactions. They must protect this information against unauthorized access, data breaches and cyberattacks.

Regulatory Compliance and Auditing:

Regulatory compliance means following applicable laws, standards and industry requirements, governing data

protection, privacy, security and financial operations.

Auditing involves examining systems, configurations, access records and security procedures.

Role of Logging and Monitoring:

Logging records events generated by applications, Operating systems, databases and cloud services. Monitoring analyzes system activity to identify unusual behaviour, operational problems and potential security threats.

Role of Encryption:

Encryption protects confidential information by making it unreadable without the appropriate cryptographic keys.

Real World Example: Online Bank Transactions

Consider a bank that processes online fund transfers. When a customer initiates a transaction, the application communicates with the banking backend through encrypted connections.

Conclusion:

Cloud security and compliance require a combination of provider

capabilities and organizational responsibility.

5. Cost Advantages of Cloud Storage:

cloud storage can reduce costs compared with traditional storage systems in several ways.

Containers and Virtual Machines.

A virtual machine (VM) uses a hypervisor to run a virtualized hardware environment with its own guest operating system.

Real World Example:

Consider an e-commerce application consisting of separate services for user authentication, product catalogues, payments and order processing.

Conclusion:

Containers provide lightweight application packaging, efficient resource usage and portability.

Kubernetes simplifies container deployment, scaling, recovery and updates.